



Bulletin 365

Multi-Speed Drum Switches

June 1, 1960

Descriptive
Bulletin **365**



Left to right: Size 2, 0 and 1 drum switches.

Convenient manual control for all kinds of multi-speed squirrel cage motors up to 15 HP, 220/440/550 V.

Each drum is highly versatile—can be adapted for many different switching combinations.

Several interlocking arrangements available.

Rugged, efficient contacts make and break with minimum arcing, give long trouble-free service.

Wrap-around covers, generous wiring space, easy-to-reach terminals and conveniently placed conduit knockouts—these features all help to make installation quicker and easier.

Compact enclosures save valuable space, stand up under abuse, look good with any machine.

Application

Bulletin 365 drum switches are designed for manual starting, speed changing and reversing of multi-speed squirrel cage motors rated up to 15 HP, 220/440/550 volts. The motor can be separate winding or consequent pole; if consequent pole, it can be variable torque, constant torque or constant horsepower.

Compact, rugged and highly versatile, these drum switches are frequently used to control mixers, compressors, tumblers, blowers, fans, pumps, special machine tools and many other types of equipment that are driven by multi-speed motors and require manual regulation. Bulletin 365's can be furnished to provide either two, three or four speeds for non-reversing, full reversing or single speed reversing applications.

Drums with ratings up to and including 10 HP are suitable for switching the motor circuit directly on the switch contacts, but those with higher ratings require a magnetic contactor or a similar device to switch the motor circuit.

Operation

The speed of a multi-speed squirrel cage motor is changed by increasing or decreasing the number of poles. With consequent pole motors, the Bulletin 365 does this by regrouping poles; that is, by reconnecting alternate sections of the same winding. The 365 changes the speed of separate winding motors by connecting individual (separate) windings or groups of windings to give the various speeds. For reverse operation of either kind of motor, the drum switch connects the windings as for forward operation but transposes two of the line connections.

Design

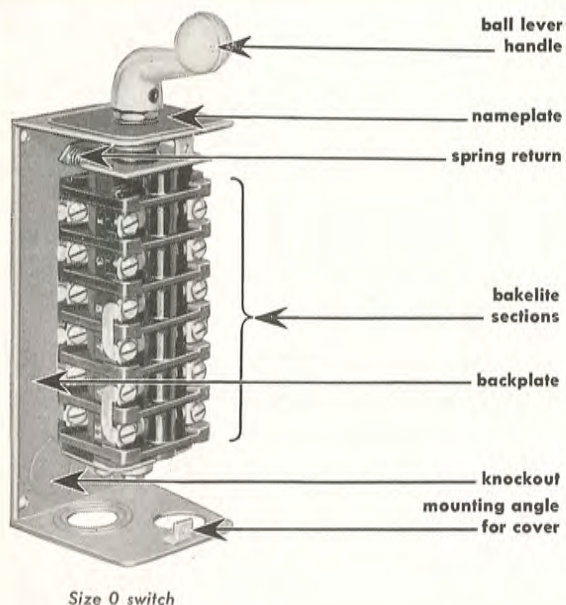
Sizes 00 and 0—These spring opened, cam closed drums are for motors rated up to 2 HP, 440/550 volts. The main components are labelled in the illustrations at the top of Page 2.

The operating mechanism in these sizes is a group of individual sections bolted together to form a solid assembly. Each section consists of a bakelite molding carrying two stationary and two movable contacts. Molded cams cause the contacts to close by forcing the movable contacts against the stationary contacts. This action places the movable contact under spring tension, which opens the contacts when the cam pressure is removed.

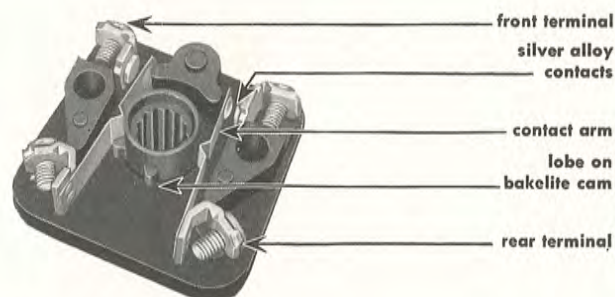
The cams have double hexagonal holes and are mounted on a hexagonal shaft. Such a construction eliminates the possibility of relative motion between the shaft and cam, and at the same time permits mounting the cam in any one of 12 positions on the shaft. It is therefore possible to obtain a very large number of switching arrangements.

The butt type contacts, stationary and movable, are made of cadmium silver alloy, a material which provides excellent connections with a minimum of arcing. Since the oxides which form on the alloy conduct as efficiently as the alloy itself, the contacts give long, uninterrupted service, never need cleaning, filing or dressing. No contact adjustment is necessary, either; the spring tension in the contact fingers compensates for normal wear. Any arcing that might occur is confined by the bakelite sections, which act as arc barriers.

Normally Size 00 switches are furnished without a pilot interlock, but various types of interlocks can be sup-



Details of Construction Sizes 00, 0



Bakelite section from Size 0 switch

plied for a small additional charge. Standard Size 0 switches have a Type R interlock, which can be replaced by a Type PP interlock at extra cost if the customer so desires. Interlocks are described on Page 3.

The standard operating handle for Size 00 and 0 drums is the ball lever type, but a pistol grip, tear drop or knurled knob handle can be substituted at no extra cost.

Size 1—Size 1 switches are for motors rated from 3 HP, 110 volts to 7½ HP, 440/550 volts. The major parts of this drum are identified at the top of Page 3.

The Size 1 switch is similar in appearance to Sizes 00 and 0 but is considerably larger. It has many of the same features found in the smaller switches, including molded bakelite sections and cams, butt type silver alloy contacts and flat spring contact fingers.

However, the Size 1 is cam opened and spring closed—just the opposite of the Size 00 and 0 drums. The contacts are opened by the tension of coil springs attached to the contact fingers and the base of the section. A metal shield surrounds the contacts and prevents the arc from burning the adjacent bakelite. A molded projection keeps connecting wires away from the contacts.

Size 1 switches are supplied with a standard Type R interlock, but for a small additional charge the optional Type PP can be substituted if the customer requests it. A ball lever handle is standard on all Size 1 drums.

Size 2—The construction of these larger drums, which are designed to control motors having a maximum rating of 15 HP, 220/440/550 volts, differs considerably from that of the Size 00, 0 and 1 switches. Here the stationary contacts are part of finger assemblies fastened to a bakelite "finger board", and the movable contacts are mounted on a cylinder attached to the shaft. When the operating lever is turned, the shaft rotates and causes certain station-

ary contacts to make connections with the movable contacts for the desired speed and direction.

Contact fingers are non-stubbing, self-aligning and adjustable for wear. Each assembly consists of a finger support, pressure spring, flexible shunt, terminal lug, contact tip and movable finger.

The contact tips are made of hard extruded copper and are plated with silver to assure good electrical contact with the shunt, which forms a low resistance path from the tip to the terminal lug. Made of laminated tinned copper, the shunt lies on top of the finger, where it cannot be damaged by the arc. Motor current is kept away from the contact finger, so that the spring remains cool and keeps its temper indefinitely.

The movable contacts are mounted on semi-cylindrical bakelite segments bolted to the shaft. The segments can be removed from the front of the drum without disturbing the wiring.

All these large drums are operated by means of a hand lever. In the Size 2 switches for three and four speed reversing service, the lever is built so the operator must press down on the grip before he can turn the lever.

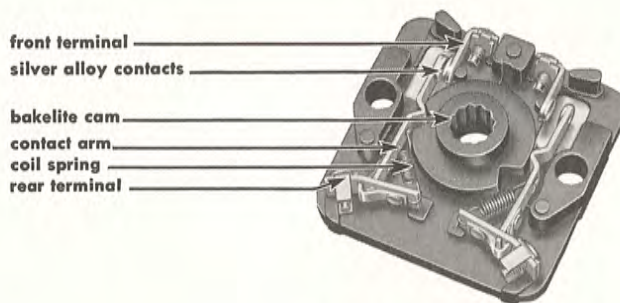
Size 2 drums when used with motors rated more than 10 hp must be used with a magnetic contactor or similar automatic device. This magnetic switch, which provides no-voltage and undervoltage protection, can be controlled by means of a separate pilot device or the "Start-Stop" push button station built into the top plate of each drum.

Interlocks . . .

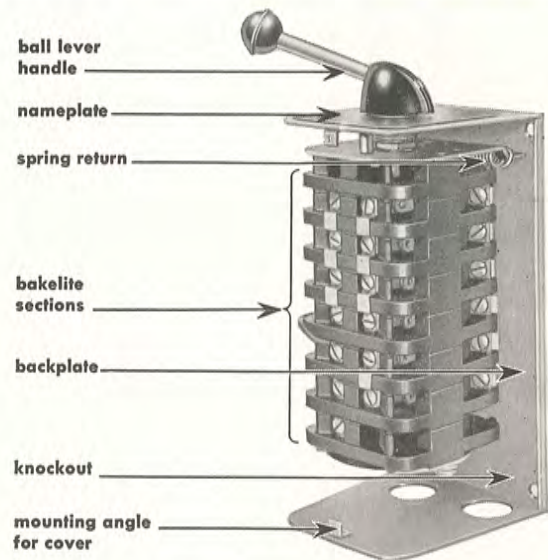
Type R—Standard on all Size 0 and 1 switches, the Type R interlock provides "Off" position reset (Size 00 switches are not normally furnished with an interlock). The

Details of Construction

Size 1



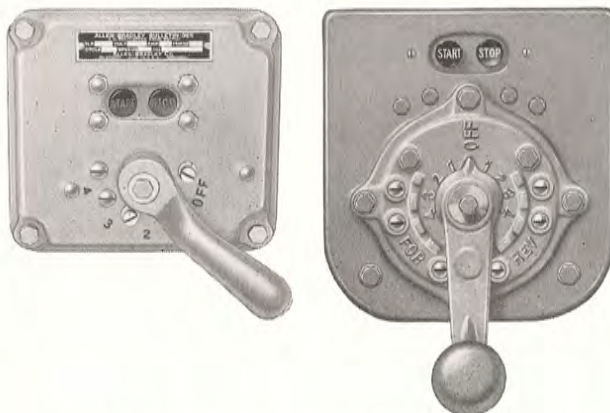
Bakelite section from Size 1 switch



Size 1 switch

drum lever must be in the "Off" position before the auxiliary magnetic contactor or automatic switch can be closed, and the motor must always be started on low speed. The circuit is made and opened at the drum contacts when changing speeds.

Type W—This interlock is standard on all Size 2 drums. Every time the drum lever is moved, the Type W interlock opens the pilot circuit of the magnetic contactor or automatic switch, thus providing no-voltage protection for all speeds in either direction. After the drum is set for the desired speed, the motor can be started and stopped by means of the "Start-Stop" push buttons on the top plate or by a separate push button station.



Top plates for the larger drum switches. The plate shown at left is standard on Size 2 switches for two speed reversing and three and four speed non-reversing applications. The plate at right is used on Size 2 three and four speed reversing switches. The lever used with the latter type of plate has a spring loaded grip which must be depressed before the position of the lever can be changed. Note that both plates have built-in "Start-Stop" push button stations for controlling an auxiliary magnetic contactor or automatic starter.

Type PP—The Type PP is a push-pull interlock which provides the equivalent of three wire push button control. For a small additional charge, it can be furnished on Size 0 and 1 drum switches instead of the standard Type R interlock.

Drums having the Type PP interlock are arranged so that the operator must press down on the handle before he can move it to another position. Depressing the handle opens the magnetic switch, which recloses as soon as the handle is allowed to return to its normal position.

Enclosures

Sizes 00, 0 and 1—The standard enclosure for these three sizes is a Nema Type 1 enclosure consisting of a channel-shaped backplate and a wrap-around cover. Removing the cover exposes three sides of the switch mechanism, including the terminals and contacts. This construction not only facilitates wiring but also permits a thorough inspection without having to remove any parts.

Sturdy and modern in appearance, the whole enclosure is made of sheet steel finished with baked gray enamel. It has four knockouts, two in the bottom, two in the back, and is drilled for surface mounting. Though the external dimensions are small, there is ample wiring space inside.

Size 00, 0 and 1 drums are suitable for either surface mounting or for flush mounting on a panel up to 1/4" thick. They can also be modified for cavity mounting by adding an escutcheon plate; the extra charge for this plate is listed on Price Sheet 365. If specified on the order, Size 00, 0 and 1 drums can be supplied without an enclosure or without the wrap-around cover.

Size 00 and 0 switches can be specially mounted in a Nema Type 4 enclosure. For prices and information, see your Allen-Bradley representative.

Size 2—The enclosures for these larger drums have cast iron plates at the top and bottom supported by a sheet steel plate in back and two L-shaped braces in front. It is exceptionally easy to wire and inspect the drums because, when the wrap-around cover is removed, two sides and the front of the switching mechanism are exposed. All enclosure surfaces, cast iron and sheet steel, are finished with a baked gray enamel.

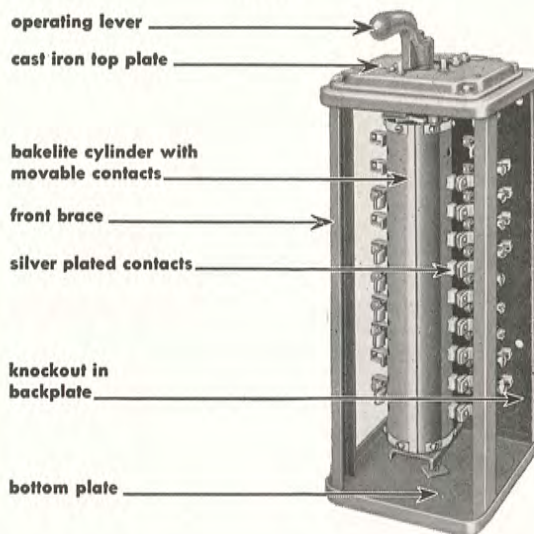
Size 2 switches can be supplied without the enclosure or without the wrap-around cover. To facilitate wiring, a conduit box can be supplied at the bottom of the enclosure. This feature is available at extra cost.

Information Required with Orders • • •

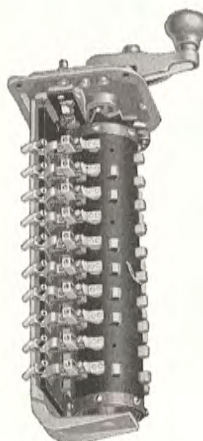
1. Line voltage.
2. Horsepower rating at each speed.
3. Phase.
4. Type of motor—separate winding or consequent pole. If consequent pole, specify either variable torque, constant torque or constant horsepower.
5. State whether drum switch is to be non-reversing, reversing on one speed or reversing on all speeds. If reversing on one speed, designate the speed to be reversed.
6. Number of speeds in both directions.
7. Revolutions per minute at each speed.
8. Furnish diagrams of internal motor connections. Wiring Diagrams 365 contains illustrations of typical connections.

Details of Construction

Size 2



Size 2 switch



Size 2 drum switch without enclosure. In this construction, the movable contacts are mounted on two semi-cylindrical bakelite sections. The stationary contact finger assemblies are attached to a bakelite "finger board", which is fastened to the vertical part of the L-shaped drum frame. This drum switch mechanism can either be equipped with a standard enclosure or mounted in a machine frame.